

Module 1: Cybersecurity Overview



Topic 2: Attacks and Vulnerabilities



Cyber Attacks - Vulnerabilities - Exploits - Threats

Vulnerability: Weakness or flaw in the system that can be used by an attacker.

Threat: An activity intentionally or unintentionally with potential to cause damage to system or data.

Exploit: Piece of code or program that can allow attacker to get access to vulnerable system through a vulnerability.

Cyber Attack: Malicious activity that tries to perform unauthorized action such as collecting, disrupting, deny access or destroying on a system or data.

Cyber Attacks - Vulnerabilities - Exploits - Threats

Example 1

Vulnerability: Weak lock

Threat: Robbery

Exploit: Metal tool used by thief



Cyber Attacks - Vulnerabilities - Exploits - Threats

Example 2

Vulnerability: computer without antivirus

Threat: Data stolen

Exploit: Virus



Types of Vulnerabilities

Network Based	Flaws due to bad network design or incorrect architecture that would allow attacker to attack the network and all systems
System Based	Flaws in operating system itself that could allow attack to get access or steal data
Application based	Flaws in the application (desktop/ web/ mobile) that allows attacker to compromise hosting system
People based	Flaws that happens due to people practices or people are reason for it.

Types of Vulnerabilities

- **Network based vulnerabilities**

Ex: network without firewall allowing access from outside to inside

- **System based vulnerabilities**

Ex: Operating system weaknesses/ System misconfiguration

- **Application based vulnerabilities**

Ex: Outdated application, bad configurations

- **People based vulnerabilities**

Ex: Weak credentials, Social Engineering

Types of Exploits

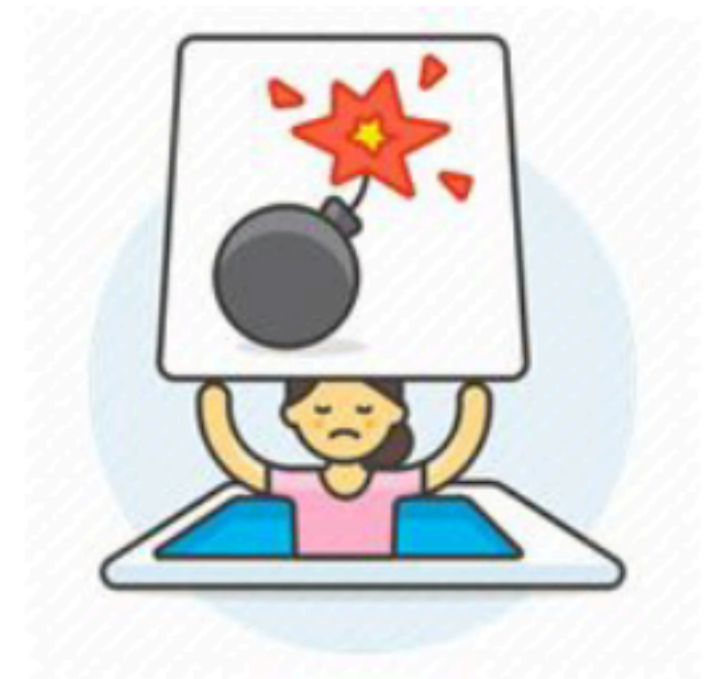
- **Server Side Exploit**

Local : Exploit require access to system and usually to increase privileges

Remote: Exploit works over network and doesn't require access to system

- **Client Side Exploit**

Exploit User App and may require user interaction thus may be combined with social Engineering



Examples of Computer Attacks



Ransomware

Ransomware is malicious software that infects your computer and displays messages demanding a fee to be paid in order for your system to work again.

Note: Ransomware is could infect endpoints and servers.

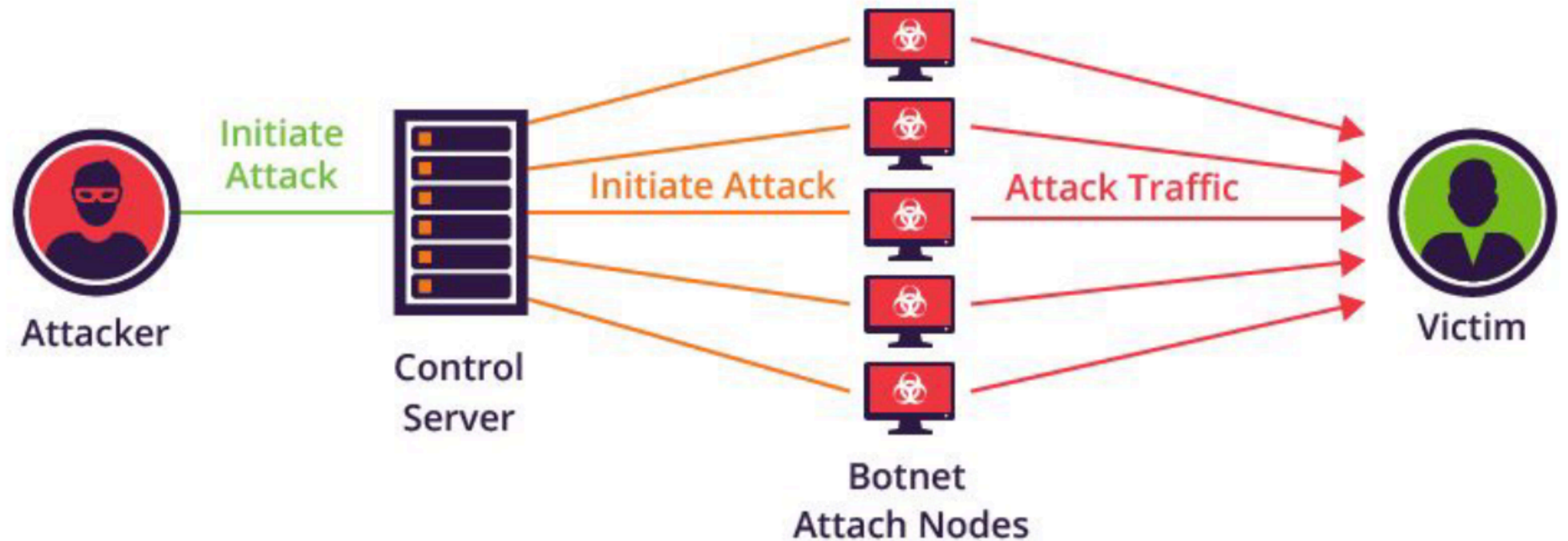
Ransomware



DDOS

- DDOS: distributed denial-of-service-attack.
- A way to overwhelm servers with requests from different devices to prevent authorized users from using their system.
- Using thousands of devices to send requests to victim may cause server break down.
- Usually happens using botnet: a numerous of infected devices by malware that are also connected by threat actor.

DDOS: Botnet



Phishing

- Phishing is a type of social engineering where an attacker sends a fraudulent (e.g. spoofed, fake, or otherwise deceptive) message designed to trick a person into revealing sensitive information to the attacker[1] or to deploy malicious software on the victim's infrastructure like ransomware.



How to Prevent Cyber Attacks?

- Use firewalls in your network
- Ensure your programs are up to date
- Use strong passwords with capital, small letters mixed with numbers and symbols (ex: "\$pr1nTs - T4l3n75)
- Avoid opening untrusted or unknown links or websites
- Avoid opening untrusted or unknown attachments
- Use good and trusted antivirus on your device
- Avoid connecting to unknown or untrusted wireless networks